

EFX₀ Allocations with at Most Three Relevant Goods: A Short Proof by Exchange Cycles

Long version

Evan Lin

University of Illinois Urbana-Champaign

Abstract. An allocation of indivisible goods is EFX₀ if no agent envies another agent’s bundle after removing any single good from it, even a good that the envious agent values at zero. We give a short proof that every instance with nonnegative additive valuations in which each agent positively values at most three goods has a complete EFX₀ allocation in which all bundles but at most one have at most two goods, and that one can be computed in polynomial time by an algorithm we call Draft and Exchange (DE). DE lets agents that can leave with one good do so, drafts one good each for the others, and then, while no agent can absorb the leftover goods, exchanges goods along a cycle or a path of agents, each of which gets strictly better off; there are at most $4n$ exchanges. The heart of the proof is an *Improvement Lemma*: a valid state from which no free agent can absorb the leftovers is Pareto-dominated by another valid state, obtained by one such exchange. The proof, derived independently twice and refereed once with no error found (by an independent AI agent; there has been no human peer review), is machine-checked in Lean, together with the algorithm, its bound of $4n$ exchanges and its running time $O(n(n + m))$; it was also checked by computer on millions of states.

Keywords: Fair division · Indivisible goods · EFX · Serial dictatorship · Exchange cycles

1 Introduction

1.1 EFX and EFX₀

We allocate a set M of m indivisible goods to a set N of $n \geq 1$ agents. Agent i has an *additive valuation*: a value $v_i(g) \geq 0$ for each good g , and $v_i(S) = \sum_{g \in S} v_i(g)$ for every set S of goods. An *allocation* X assigns every good to exactly one agent; agent i ’s *bundle* X_i is the set of goods assigned to i . So the bundles are pairwise disjoint and cover M : every allocation in this paper is *complete*, and no good may be thrown away.

Envy-freeness can be impossible with indivisible goods (one good, two agents who value it). *Envy-freeness up to any good* weakens it [4]: the envy

of i for X_j must disappear as soon as any single good is removed from X_j . Two versions differ in which goods may be removed:

EFX: $v_i(X_i) \geq v_i(X_j \setminus \{h\})$ for all $i \neq j$ and $h \in X_j$ with $v_i(h) > 0$;

EFX₀: $v_i(X_i) \geq v_i(X_j \setminus \{h\})$ for all $i \neq j$ and $h \in X_j$.

Both conditions are vacuous when X_j is empty. EFX₀ implies EFX, and the two coincide when all values are positive. Whether every instance with additive valuations has a complete EFX allocation is a central open question of fair division [4]. The difference between the two notions is one rule: *under EFX₀, if another agent's bundle contains a good that agent i values at 0, then i must not envy that bundle at all*, because removing that good does not lower its value to i .

Example 1 (EFX, but not EFX₀). Two agents share three goods a, b, c . Agent 1 values a at 3, b at 2 and c at 0; agent 2 values only c , at 1. Consider $X_1 = \{b\}$ and $X_2 = \{a, c\}$. Under EFX, agent 1 may remove only a from X_2 , since c is worthless to it; what remains, $\{c\}$, is worth $0 \leq 2$. Agent 2 holds c and faces the single good b , which it values at 0. So X is EFX. It is not EFX₀: removing c from X_2 leaves $\{a\}$, worth $3 > 2$ to agent 1. The allocation $X_1 = \{a\}$, $X_2 = \{b, c\}$ is EFX₀.

1.2 Results

Write $R_i = \{g \in M : v_i(g) > 0\}$ for the goods agent i values, its *relevant* goods. For $k \leq 2$, if every agent values at most k goods, serial dictatorship works: the agents, in any order, each take a favourite remaining good, and the last agent takes everything left; the result is EFX₀ ([11]; Corollary 1 below). At three goods this breaks. An agent that values a, b, c at 4, 3, 2 may take a and then see b and c end up in the last agent's bundle together with a third good; removing that third good, which it values at zero, leaves b and c , worth $5 > 4$.

Theorem 1 (existence). *Every instance with $n \geq 1$ agents and non-negative additive valuations in which every agent positively values at most three goods has a complete EFX₀ allocation.*

Theorem 2 (one large bundle). *Every such instance has a complete EFX₀ allocation in which all bundles but at most one have at most two goods. In particular this holds when every agent values exactly three goods and is balanced, that is, values none of them more than the other two together.*

Theorem 3 (algorithm). *Algorithm Draft and Exchange (DE, Section 6) computes, for every instance in which every agent positively values at most three goods, an allocation as in Theorem 2. It performs at most n peeling steps and at most $4n$ exchanges, and each step takes time polynomial in n and m . On real values we count a comparison of two sums of at most two values of one agent as one step; every decision of DE is such a comparison.*

DE has three stages: *peeling* (agents that can leave with one good do so), a *draft* (the remaining agents take at most one good each), and *exchanges* (goods move along cycles or paths of agents, each of which gets strictly better off), after which one agent, the *absorber*, takes the leftover goods. Section 2 explains them.

1.3 The proof, and its status

The core of the proof is the Improvement Lemma (Theorem 5): in a *valid* state (Section 4) from which no free agent can absorb the leftovers, a single exchange along a cycle or a path gives a valid state in which every agent is at least as well off and some agent is strictly better off. Since utilities only grow, exchanges stop; when they stop, an absorber exists, and the completion is EFX₀ by a soundness theorem that needs nothing but validity (Theorem 4).

Status. We are explicit about what has been verified, and how. The Improvement Lemma, the soundness theorem in the form used here, the algorithm DE, its correctness, and Theorem 2 for all instances with at most three relevant goods per agent are written proofs, and they are *machine-checked in Lean 4*, also for ordered values such as the nonnegative reals (Section 8), as are the other results of this paper and the running time of DE, with the exceptions listed in Section 8. The Improvement Lemma was derived independently twice, by two different arguments (Section 5.5). Its proof was refereed once, with no error found, by an independent AI agent (a separate session of the coding assistant that helped write this work, which did not write the proof); there has been no human peer review. We apply the referee’s four presentation fixes. Every step of it has been checked by computer as described in Section 8: three independent implementations, two of this proof and one of the second, assert every claim on millions of states.

Table 2 in Section 8 lists every claim with its evidence. Section 2 explains the approach informally; Sections 3–6 give the definitions, the

proofs and the algorithm; Section 7 shows that the shape of Theorem 2 cannot be improved in the obvious ways. The notes from which this paper is written, the test programs and the referee report are in the repository [10].

1.4 Related work

For additive valuations, complete EFX allocations are known to exist only in special cases, among them: for three agents [5] and, more generally, for three types of agents (at most three distinct valuations) [7]; when $m \leq n + 3$ [13]; for four agents and at most nine goods [3]; when every good is valued by at most two agents (multigraphs of agents, with goods as edges) [1], with special cases in [6,14,8]; and for hypergraphs of girth at least four [9]. All of these use the all-goods form EFX_0 . For monotone valuations that are not additive, EFX allocations need not exist [2,12]. Table 1 compares these results with ours.

We restrict instead how many goods each agent values. For at most two goods per agent, serial dictatorship suffices and has been formalized in Lean [11]. For at most four, Viswanathan and Mehta [15] announce the existence of ordinary EFX allocations, with a polynomial-time algorithm and a proof sketch in an extended abstract. Their first phase, which gives a first choice, by a maximum matching, to as many as possible of the agents that value their top good more than their second and third together, resembles our peeling rule R1. Their algorithm ends by giving the goods that nobody values to an arbitrary agent, which is harmless for EFX but can break EFX_0 , as Example 1 illustrates. Since EFX_0 implies EFX, Theorem 1 proves the case of three goods of their result in full.

An instance with at most three relevant goods per agent and no complete EFX_0 allocation would refute the additive EFX conjecture (perturb all values by a small $\varepsilon > 0$; compare [5]), but perturbing makes every good relevant to every agent, and no result we know of implies Theorem 1.

2 Overview

This section explains the approach in plain words; everything is defined and proved later.

Peeling. If agent i values its favourite remaining good p at least as much as all its other remaining goods together, it may take $\{p\}$ and leave: it values every other bundle at most at $v_i(p)$, and nobody strongly envies a

Table 1. Existence results for complete allocations under restrictions. *Version cited:* the published version where one exists, else arXiv. ^aGoods nobody values go to an arbitrary agent. ^bIncludes additive. ^cExtended abstract with a proof sketch. ^dAn AAMAS 2025 extended abstract has the degree and girth conditions. ^eAccepted at EC 2026.

Result	Restriction (agents, goods, who values what)	Notion	Complete	Valuations	Algorithm	Version cited
[5]	$n = 3$	EFX ₀	yes	additive	pseudo-polynomial	JACM 2024
[7]	at most three distinct valuations	EFX ₀	yes	additive	no bound stated	EC 2025
[13]	$m \leq n + 3$	EFX ₀	yes	monotone	none stated	MOR 2024
[3]	$n = 4, m \leq 9$	EFX ₀	yes	additive	none stated	arXiv
[6]	graphs: at most two valuers per good, two agents share at most one good	EFX ₀	yes	monotone	polynomial time	MOR 2026 (EC 2023)
[1]	multigraphs: at most two valuers per good	EFX ₀	yes	cancelable ^b	polynomial time	arXiv ^e
[14]	multigraphs: bipartite, or at most $\lceil n/4 \rceil - 1$ neighbours per agent, or girth at least six (non-parallel edges)	EFX ₀	yes	monotone	not efficient (stated)	arXiv ^d
[8]	at most two valuers per good, $v_i(g) \in \{0, v_g\}$	EFX ₀	yes	restricted additive	no time bound stated	WINE 2024
[9]	hypergraphs of girth at least four (a good joins its valuers)	EFX ₀	yes	monotone	polynomial time	AAAI 2026
[15]	at most four relevant goods per agent	EFX	yes ^a	additive	polynomial time, announced	AAMAS 2024 ^c
[11]	at most two relevant goods per agent	EFX ₀	yes	additive; monotone	serial dictatorship; Lean-checked	GitHub, v1.2.0
this paper	at most three relevant goods per agent; any number of valuers per good	EFX ₀	yes	additive	polynomial time, at most $4n$ exchanges	this paper

bundle of one good (Lemma 3). We repeat this (*rule R1*). When nobody can leave, every remaining agent values exactly three remaining goods $a \succ b \succ c$, with $v(a) < v(b) + v(c)$ (Lemma 4): the *core case*.

States. In the core case, a *state* lets each agent hold nothing, one of its goods, or its pair $\{b, c\}$; the other goods are *leftover*. Utilities are ordinal: pair, then a , b , c , then nothing. An agent holding its pair is always safe. An agent holding one good is safe as long as the goods it ranks higher, its *needs*, stay alone in their bundles, with one exception: an agent holding only a is not safe if b and c lie together in a bundle of three or more goods (Lemma 2). A state is *valid* if every needed good is held alone. To finish, one agent, the *absorber*, takes the leftovers. It must hold no needed good, and every agent x that holds only its top and would find b_x and c_x in the absorber’s bundle (an *exposed* agent) must be protected by sending one of them to another free agent, one good each. If this is possible the state is *completable*, and the result is EFX_0 (Theorem 4).

Improving. The draft gives a valid state, which may not be completable. The Improvement Lemma (Theorem 5) says that then a valid state Pareto-dominates it, by a *need cycle* (agents in a cycle take the good of the previous one, which they need), a *pair chain* (an agent whose b and c are both leftover takes them, and its top travels along agents that need it), or an *exchange cycle*, which mixes need arcs with *exposure arcs* $o \rightarrow x$: x , exposed for the free agent o , takes its pair, made of o ’s good and a leftover good. The key step is a count. If no free agent can absorb, each free agent has at least $|F|$ distinct protecting goods, where F is the set of free agents, so distinct ones can be chosen for all free agents greedily. Each free agent then points to an exposed agent, every other agent to an agent that needs its good, and this map has a cycle. In the instance of Section 6.3, every improvement gives pairs to two agents at once; a single *rotation*, an exchange with one exposure arc, is not enough there (Section 6.4).

3 Preliminaries

3.1 Model, threats and safety

An *instance* consists of a finite set N of $n \geq 1$ agents, a finite set M of m goods, and nonnegative additive valuations v_i , $i \in N$. Agents and goods are numbered, and “index order” and “the first” refer to these numbers.

For a set B of goods, the *threat* of B to agent i is

$$\theta_i(B) = v_i(B) - \min_{h \in B} v_i(h) \quad \text{if } B \neq \emptyset, \quad \theta_i(\emptyset) = 0.$$

By additivity, $\theta_i(B)$ is the largest value of $v_i(B \setminus \{h\})$ over $h \in B$. Agent i is *safe* in an allocation X if $v_i(X_i) \geq \theta_i(X_j)$ for every agent $j \neq i$. So X is EFX₀ if and only if every agent is safe. A good is *alone* in X if it is the only good of its bundle.

Lemma 1 (threats). *For every agent i and every set B of goods: (a) if $|B| \leq 1$, then $\theta_i(B) = 0$; (b) $\theta_i(B) \leq v_i(B \cap R_i)$; (c) $\theta_i(\{g, h\}) = \max(v_i(g), v_i(h))$ for goods $g \neq h$.*

Proof. (a) $\theta_i(\{h\}) = v_i(h) - v_i(h) = 0$. (b) The minimum is at least 0, and $v_i(B) = v_i(B \cap R_i)$ because the other goods of B are worth 0 to i . (c) $v_i(g) + v_i(h) - \min(v_i(g), v_i(h)) = \max(v_i(g), v_i(h))$. $\square$

When $|R_i| = 3$ we name agent i 's relevant goods a_i, b_i, c_i so that $v_i(a_i) \geq v_i(b_i) \geq v_i(c_i) > 0$, breaking ties between equal values by index. This gives a strict *ranking* $a_i \succ_i b_i \succ_i c_i$, fixed once and for all; $g \succ_i h$ implies $v_i(g) \geq v_i(h)$. Agent i is *balanced* if $v_i(a_i) \leq v_i(b_i) + v_i(c_i)$, and *strictly balanced* if $v_i(a_i) < v_i(b_i) + v_i(c_i)$.

The next lemma is the half of the description of safety that the proofs need: it says when an agent with three relevant goods is certainly safe. (Section 7 gives the full description, which only the counterexamples there use.)

Lemma 2 (safety). *Let X be an allocation and i a balanced agent with $|R_i| = 3$.*

- (a) *If $b_i, c_i \in X_i$, then i is safe.*
- (b) *Let $y \in X_i \cap R_i$, or $y = \perp$ ("no good"). Let $G_y = \{g \in R_i : g \succ_i y\}$, and $G_\perp = R_i$. Suppose that no bundle X_j with $j \neq i$ and $|X_j| \geq 2$ contains a good of G_y . Then i is safe, unless $y = a_i$ and some bundle X_j with $j \neq i$ and $|X_j| \geq 3$ contains both b_i and c_i .*

Proof. Fix $j \neq i$. If $|X_j| \leq 1$, then $\theta_i(X_j) = 0$ by Lemma 1(a). So let $|X_j| \geq 2$, and let $T = X_j \cap R_i$; then $\theta_i(X_j) \leq v_i(T)$ by Lemma 1(b).

(a) X_j is disjoint from X_i , which contains b_i and c_i , so $T \subseteq \{a_i\}$ and $v_i(T) \leq v_i(a_i) \leq v_i(b_i) + v_i(c_i) \leq v_i(X_i)$.

(b) If $y = \perp$, then $T \subseteq R_i = G_\perp$, so $T = \emptyset$ by hypothesis, and $\theta_i(X_j) \leq 0$. Otherwise $y \in X_i$. A good $g \in T$ is not in G_y (by hypothesis)

and is not y (which lies in X_i , not in X_j), so $y \succ_i g$ and $v_i(g) \leq v_i(y)$. If $|T| \leq 1$, then $v_i(T) \leq v_i(y) \leq v_i(X_i)$. If $|T| \geq 2$, then y and two goods ranked below it are three goods of R_i ; as $|R_i| = 3$, this forces $y = a_i$ and $T = \{b_i, c_i\}$. If $|X_j| = 2$, then $X_j = \{b_i, c_i\}$ and $\theta_i(X_j) = v_i(b_i) \leq v_i(a_i) \leq v_i(X_i)$ by Lemma 1(c). If $|X_j| \geq 3$, we are in the exception. $\square$

3.2 Peeling

Lemma 3 (peeling). *Let $i \in N$ and $P \subseteq M$ satisfy (i) $v_i(P) \geq v_i(M \setminus P)$, and (ii) $|P| \leq 1$. If the instance with agents $N \setminus \{i\}$ and goods $M \setminus P$ has an EFX_0 allocation Z , then Z together with $X_i = P$ is an EFX_0 allocation of the whole instance.*

Proof. Agent i : every other bundle Z_j lies in $M \setminus P$, so $\theta_i(Z_j) \leq v_i(Z_j) \leq v_i(M \setminus P) \leq v_i(P)$, using that the minimum in the threat is nonnegative, and (i). An agent $j \neq i$ towards P : $\theta_j(P) = 0$ by (ii) and Lemma 1(a). An agent $j \neq i$ towards Z_k with $k \neq i, j$: this is a constraint of Z , and j still holds Z_j . $\square$

Rule R1. Let G be the set of remaining goods, and let p be agent i 's favourite good of G (largest value, ties by index). If $v_i(p) = 0$, so that i values no good of G , then i may leave with $P = \emptyset$, and (i) reads $0 \geq 0$. If $v_i(p) > 0$ and $v_i(G \setminus \{p\}) \leq v_i(p)$, then i may leave with $P = \{p\}$. Rule R1 applies, for instance, to every agent that values at most two remaining goods, and to every agent with $v_i(a_i) \geq v_i(b_i) + v_i(c_i)$.

Corollary 1 (two relevant goods). *If $|R_i| \leq 2$ for every agent, then serial dictatorship, in any order of the agents, each taking a favourite remaining good (or nothing once no good is left), with the last agent taking all remaining goods, returns an EFX_0 allocation.*

Proof. By induction on the number of agents. One agent faces no constraint. Otherwise the first agent i values at most two remaining goods, so rule R1 applies to its choice: its other relevant good, if any, is worth at most the good it takes. The rest of the procedure is serial dictatorship on the smaller instance, which is EFX_0 by induction, and Lemma 3 finishes the proof. $\square$

Lemma 4 (when R1 applies to nobody). *Let agent i value at most three goods of the set G of remaining goods, and suppose that rule R1 does not apply to i . Then i values exactly three goods of G and is strictly balanced with respect to them.*

Proof. Rule R1 fails, so $v_i(p) > 0$ and $v_i(G \setminus \{p\}) > v_i(p)$ for the favourite p . If i valued at most two goods of G , then $v_i(G \setminus \{p\})$ would be the value of at most one good, which is at most $v_i(p)$. So i values exactly three goods of G , p is its top a_i , and $v_i(b_i) + v_i(c_i) = v_i(G \setminus \{p\}) > v_i(a_i)$. $\square$

4 States and Soundness

In this section and the next we are in the *core case*: every agent values exactly three goods and is strictly balanced. (Strict balance is not used below: the order of utilities is a bookkeeping device, and soundness needs only balance.) Goods that no agent values may exist. In examples, $g_0, g_1, \dots$ are goods.

4.1 Definitions

Definition 1 (states). A state Y gives each agent i a holding Y_i , one of $\emptyset$, $\{a_i\}$, $\{b_i\}$, $\{c_i\}$ and $\{b_i, c_i\}$ (its pair), with the holdings pairwise disjoint.

- $U = \{i : Y_i = \{b_i, c_i\}\}$ is the set of pair holders.
- $J = M \setminus \bigcup_i Y_i$ is the set of leftover goods (junk).
- For $i \notin U$ with $|Y_i| = 1$, write $Y_i = \{y_i\}$.
- The utility of i is $u_i(Y) = 4, 3, 2, 1, 0$ if Y_i is its pair, $\{a_i\}$, $\{b_i\}$, $\{c_i\}$, $\emptyset$. Since $v_i(a_i) < v_i(b_i) + v_i(c_i)$, a larger utility never means a smaller value.
- A state Y' Pareto-dominates Y if $u_i(Y') \geq u_i(Y)$ for every i , with strict inequality for some i .

Definition 2 (needs, validity, free agents). Let Y be a state.

- The needs of agent i are: $N_i = \emptyset$ if $i \in U$; $N_i = R_i$ if $Y_i = \emptyset$; and $N_i = \{g \in R_i : g \succ_i y_i\}$ if $Y_i = \{y_i\}$. (N without a subscript is the set of agents.) No agent needs a good it holds. The set of goods needed alone is $NA = \bigcup_i N_i$.
- Y is valid if every $g \in NA$ is the only good of some agent: $Y_j = \{g\}$ for some j . Such a j is not in U , and it is not an agent that needs g .
- An agent $i \notin U$ is free if $Y_i = \emptyset$ or $y_i \notin NA$. F is the set of free agents. An agent outside $U \cup F$ holds a single good that some agent needs.

In a valid state, the leftover goods and the goods of pair holders are not needed: every needed good is held alone.

Definition 3 (need digraph). *The need digraph D of a state has the agents outside U as vertices, and an arc $j \rightarrow j'$ (a need arc) whenever j holds a single good y_j and $y_j \in N_{j'}$. It has no loops, since no agent needs its own good.*

Definition 4 (exposure, absorbers, completions). *Let Y be a state and o an agent.*

- *An agent x is exposed for o if $x \neq o$, $x \notin U$, $Y_x = \{a_x\}$, and $b_x, c_x \in J \cup Y_o$. E_o is the set of agents exposed for o .*
- *o is a valid absorber, with a set H , if $o \in F \cup U$ and (A1) $\{b_x, c_x\} \cap J \neq \emptyset$ for every $x \in E_o$, and (A2) $H \subseteq J$ meets $\{b_x, c_x\} \cap J$ for every $x \in E_o$, and $|H| \leq |F \setminus \{o\}|$. The goods of H are the protecting goods. Y is completable if some agent is a valid absorber.*
- *The completion X of Y with a valid absorber o and its set H : each good of H goes to a different free agent other than o (there are enough, by (A2)); o receives $X_o = Y_o \cup (J \setminus H)$; every other agent i keeps $X_i = Y_i$, or $X_i = Y_i \cup \{h\}$ if it received $h \in H$.*

The completion is an allocation of all goods, since M is the disjoint union of the holdings and J .

4.2 Soundness

Theorem 4 (soundness). *Let Y be a valid state, o a valid absorber with set H , and X the completion. Then X is EFX_0 , and every bundle other than X_o has at most two goods.*

Proof. Sizes. A free agent other than o holds at most one good and receives at most one good of H . A pair holder other than o holds two goods. Every other agent $j \neq o$ is outside $U \cup F$ and keeps its single good. So only X_o can have three or more goods.

(N) *Needed goods stay alone.* Let $g \in NA$. By validity, $Y_j = \{g\}$ for some agent j . Then $j \notin U$, and $j \notin F$ because g is needed. So $j \neq o$, as $o \in F \cup U$, and j receives no good of H , which goes to free agents. Hence $X_j = \{g\}$.

(OC) *The owner constraint.* No agent $x \neq o$ with $x \notin U$ and $Y_x = \{a_x\}$ has both b_x and c_x in X_o . Indeed, suppose it had. Since $X_o = Y_o \cup (J \setminus H)$, we get $b_x, c_x \in J \cup Y_o$, so $x \in E_o$. By (A2), H contains a good of $\{b_x, c_x\} \cap J$. That good went to an agent other than o , and it is not in Y_o (it is leftover), so it is not in X_o : a contradiction.

Safety. Let i be an agent; it is balanced. If $i \in U$, then $b_i, c_i \in Y_i \subseteq X_i$, and i is safe by Lemma 2(a). Let $i \notin U$. Apply Lemma 2(b) with $y = y_i$ if $Y_i = \{y_i\}$ (then $y \in Y_i \subseteq X_i$), and $y = \perp$ if $Y_i = \emptyset$. Then $G_y = N_i \subseteq NA$, so by (N) every good of G_y is alone, and no bundle of two or more goods contains one. So i is safe unless $y_i = a_i$ and some X_j with $j \neq i$ and $|X_j| \geq 3$ contains b_i and c_i . By the sizes, $j = o$; so $i \neq o$, $i \notin U$, $Y_i = \{a_i\}$ and $b_i, c_i \in X_o$, which (OC) excludes. $\square$

The proof uses only validity, the definition of a valid absorber, and that pair holders are balanced. Nothing in it depends on how the state was reached.

4.3 The draft, and two basic facts

Lemma 5 (the draft). *Let the agents, in any order, each take the first of a_i, b_i, c_i not yet taken, or nothing if all three are taken. The result is a valid state with $U = \emptyset$.*

Proof. The holdings are single goods of R_i or empty, and no good is taken twice, so this is a state, and $U = \emptyset$. Let $g \in N_i$. Then i took a good ranked below g , or nothing, so g was already taken when i chose. It was taken by an earlier agent j , as its only good, and holdings never change during the draft. So $Y_j = \{g\}$. $\square$

Lemma 6 (two basic facts). *Let Y and Y' be states, with needs N_i and N'_i .*

(F1) *If $u_i(Y') \geq u_i(Y)$, then $N'_i \subseteq N_i$.*

(F2) *If Y is valid, every agent outside $U \cup F$ has an out-arc in D .*

Proof. (F1) If Y'_i is the pair, $N'_i = \emptyset$. If $Y'_i = \{g'\}$, then $u_i(Y) \leq u_i(Y') < 4$, so Y_i is not the pair: either $Y_i = \emptyset$ and $N_i = R_i \supseteq N'_i$, or $Y_i = \{g\}$ with $g' = g$ or $g' \succ_i g$, and then every good ranked above g' is ranked above g . If $Y'_i = \emptyset$, then $u_i(Y) = 0$, so $Y_i = \emptyset$ and $N'_i = N_i$. (F2) Let $j \notin U \cup F$. Then j holds a single good $y_j \in NA$, so $y_j \in N_{j'}$ for some agent j' . We have $j' \neq j$, since no agent needs its own good, and $j' \notin U$, since pair holders need nothing. So $j \rightarrow j'$ is an arc of D . $\square$

Lemma 7 (transfer). *Let Y be valid, and let Y' be a state such that (a) $u_i(Y') \geq u_i(Y)$ for every agent i , with strict inequality for some i , and (b) every $g \in NA$ is, in Y' , the only good of some agent. Then Y' is valid and Pareto-dominates Y .*

Proof. By (a) and (F1), $NA' \subseteq NA$. By (b), every good of NA' is the only good of some agent in Y' . Domination is (a). $\square$

5 The Improvement Lemma

Theorem 5 (Improvement Lemma). *In the core case, let Y be a valid state in which no free agent is a valid absorber and some agent is not a pair holder. Then some valid state Pareto-dominates Y . It is obtained from Y by one need cycle, pair chain or exchange cycle (defined below), which can be found in polynomial time.*

If Y is not completable, then no free agent is a valid absorber, and not every agent is a pair holder (otherwise every agent is a valid absorber with $H = \emptyset$: nobody is exposed, because exposed agents are outside U). So the theorem applies to every valid state that is not completable.

Corollary 2 (Pareto-optimal states). *In the core case, every valid state that no valid state Pareto-dominates is completable, with a free absorber, unless every agent holds its pair.*

Valid states exist (Lemma 5), there are finitely many, and a valid state with the largest sum of utilities is not dominated by any valid state. So Corollary 2 and Theorem 4 already prove Theorem 1 in the core case; DE (Section 6) finds such a state by exchanges.

The proof is split into the lemmas below. Throughout, Y is valid.

5.1 Need cycles and pair chains

Lemma 8 (need cycle; pair chain). *(a) If D has a cycle, Y is dominated by a valid state. (b) If some agent $x \notin U$ holds only its top a_x , and $b_x, c_x \in J$, then Y is dominated by a valid state.*

Proof. (a) *Need cycle.* Let $j_0 \rightarrow j_1 \rightarrow \cdots \rightarrow j_{k-1} \rightarrow j_0$ be a cycle of D , with $k \geq 2$ distinct agents. Each of them is the tail of an arc, so it is outside U and holds a single good. Define Y' : each j_{t+1} takes y_{j_t} (indices modulo k), and every other agent keeps its holding. Each y_{j_t} is a good of $R_{j_{t+1}}$, so $Y'_{j_{t+1}}$ is an allowed holding. The goods of the cycle are permuted, so the holdings are disjoint. Each j_{t+1} receives a good that it needs, so it strictly improves; the other agents are unchanged. This is condition (a) of Lemma 7. For (b): a needed good g is $Y_j = \{g\}$ for some j ; if j is off the cycle, it keeps g alone, and if $j = j_t$, then j_{t+1} holds g alone. Lemma 7 applies.

(b) *Pair chain.* Walk in D : $j_0 = x$, and while j_t has an out-arc, let j_{t+1} be an out-neighbour. If the walk repeats an agent, D has a cycle, and (a) applies. Otherwise it stops at an agent j_k ($k \geq 0$) with no out-arc.

Define Y' : x takes its pair $\{b_x, c_x\}$; for $1 \leq t \leq k$, j_t takes $y_{j_{t-1}}$, where $y_{j_0} = a_x$; every other agent keeps its holding. The old good of j_k (a_x if $k = 0$), if any, becomes leftover. Every j_t with $t < k$ has an out-arc, so it holds a single good. The holdings are disjoint: b_x and c_x were leftover, and the goods $y_{j_0}, \dots, y_{j_{k-1}}$ each move one step along the path. Agent x goes from a_x to its pair, and each j_t with $t \geq 1$ receives a good it needs: (a) of Lemma 7. For (b): b_x and c_x are leftover, so they are not needed, by validity. A needed good held by an agent off the path stays alone. The good y_{j_t} with $t < k$ moves to j_{t+1} , alone. Finally, j_k is outside U ($j_0 = x \notin U$, and for $k \geq 1$, j_k needs something) and has no out-arc, so $j_k \in F$ by (F2): its old good, if any, is not needed. Lemma 7 applies. $\square$

From now on we assume

(P) no agent $x \notin U$ with $Y_x = \{a_x\}$ has both b_x and c_x in J ,

since otherwise Lemma 8(b) applies. We do not assume that D is acyclic; the cycles found below may be cycles of D .

5.2 Free agents and protecting goods

Lemma 9 (an agent holding nothing absorbs). *Assume (P). If $o \in F$ holds nothing, then $E_o = \emptyset$, and o is a valid absorber with $H = \emptyset$.*

Proof. An agent $x \in E_o$ would have $b_x, c_x \in J \cup Y_o = J$, against (P). With $E_o = \emptyset$, (A1) and (A2) hold for $H = \emptyset$. $\square$

Lemma 10 (protecting goods are forced). *Assume (P), and let $o \in F$ hold the single good y_o . Every $x \in E_o$ has $\{b_x, c_x\} = \{y_o, h_x\}$ for a leftover good $h_x \in J$, its protecting good. Let $H_o = \{h_x : x \in E_o\}$. Then o is a valid absorber if and only if $|H_o| \leq |F| - 1$, and then $H = H_o$ is a valid set.*

Proof. For $x \in E_o$, $b_x, c_x \in J \cup \{y_o\}$, and not both are in J , by (P). As $b_x \neq c_x$, exactly one of them is y_o , and the other, h_x , is in J . So (A1) holds, and $\{b_x, c_x\} \cap J = \{h_x\}$. A set $H \subseteq J$ meets all these singletons if and only if $H \supseteq H_o$. So (A2) can be met if and only if $|H_o| \leq |F \setminus \{o\}| = |F| - 1$, and then by $H = H_o$. $\square$

The protecting good h_x depends only on x : under (P), an agent x lies in E_o for at most one free agent o that holds a good. Indeed, if $x \in E_o \cap E_{o'}$ with $o \neq o'$, then $\{b_x, c_x\}$ contains y_o and $y_{o'}$, two different goods that are not leftover, so $\{b_x, c_x\} = \{y_o, y_{o'}\}$ contains no leftover good, against Lemma 10.

Lemma 11 (no free agent). *If $F = \emptyset$, then either every agent is in U , and then every agent is a valid absorber with $H = \emptyset$, or D has a cycle.*

Proof. If every agent is in U , nobody is exposed (exposed agents are outside U), so (A1) and (A2) hold with $H = \emptyset$. Otherwise D has a vertex, and every vertex is outside $U \cup F$, so it has an out-arc by (F2), to another vertex. A finite nonempty digraph in which every vertex has an out-arc has a cycle. $\square$

5.3 Exchange cycles

Definition 5 (exchange digraph). *Assume (P). The exchange digraph D^+ has the agents outside U as vertices, the need arcs of D , and an exposure arc $o \rightarrow x$ for every free agent o that holds a good and every $x \in E_o$; this arc carries the label h_x .*

The head of an exposure arc holds its top, so it needs nothing; the head of a need arc needs something. So the kind of an arc is determined by its head.

Definition 6 (exchange along a cycle). *Let $C = (w_0, w_1, \dots, w_{k-1})$ be a cycle of D^+ with distinct agents and $k \geq 2$, indices modulo k . The exchange along C is the state Y' in which, for every t :*

- *if $w_t \rightarrow w_{t+1}$ is an exposure arc, w_{t+1} takes its pair $\{b_{w_{t+1}}, c_{w_{t+1}}\} = \{y_{w_t}, h_{w_{t+1}}\}$;*
- *if $w_t \rightarrow w_{t+1}$ is a need arc, w_{t+1} takes y_{w_t} alone;*

and every agent off C keeps its holding. A need cycle is the case without exposure arcs.

Every agent of C holds a single good: the tail of a need arc holds a single good by definition, and the tail of an exposure arc is a free agent that holds a good. Each agent of C has exactly one predecessor on C , so Y' is well defined.

Lemma 12 (exchange cycle). *Assume (P). Let C be a cycle of D^+ whose exposure arcs have pairwise distinct labels. Then the exchange along C is a valid state that Pareto-dominates Y , in which exactly the heads of the exposure arcs of C become new pair holders.*

Proof. *Disjoint holdings.* Each good y_{w_t} moves to w_{t+1} only. The labels of the exposure arcs of C are leftover goods, so nobody holds them in

Y , and they are pairwise distinct by hypothesis. The agents off C keep goods that are neither leftover nor held by agents of C . So the holdings of Y' are disjoint, and each is an allowed holding: a pair, or a single good of $R_{w_{t+1}}$ (a needed good).

Improvement. The head of an exposure arc holds only its top in Y and its pair in Y' : utility 3 to 4. The head of a need arc receives a good it needs, which it ranks above its old good (or it held nothing): its utility strictly increases. The agents off C are unchanged. This is (a) of Lemma 7.

Needed goods. Let $g \in NA$, with $Y_j = \{g\}$. If $j \notin C$, then j keeps g alone. If $j = w_t$, then w_t is not free (its good is needed), so the arc $w_t \rightarrow w_{t+1}$ is a need arc, and w_{t+1} holds g alone. This is (b) of Lemma 7, which gives validity and domination. The new pair holders are the heads of the exposure arcs. $\square$

Lemma 13 (an exchange cycle exists). *Assume (P). Suppose that $F \neq \emptyset$, that no free agent holds nothing, and that $|H_o| \geq |F|$ for every $o \in F$. Then D^+ has a cycle whose exposure arcs have pairwise distinct labels; so Y is dominated by a valid state.*

Proof. Distinct representatives. List $F = \{o_1, \dots, o_f\}$. For $i = 1, \dots, f$ in turn, pick $q_{o_i} \in H_{o_i} \setminus \{q_{o_1}, \dots, q_{o_{i-1}}\}$. This is possible because $|H_{o_i}| \geq f > i - 1$. Let $x_{o_i} \in E_{o_i}$ be an agent whose protecting good is q_{o_i} . (The agents x_o are then distinct, since h_x depends only on x ; the argument below does not need this.)

A functional digraph. Define σ on the agents outside U : $\sigma(o) = x_o$ for $o \in F$, and $\sigma(j)$ is an out-neighbour of j in D for $j \notin U \cup F$, which exists by (F2). Then $\sigma(j) \notin U$ and $\sigma(j) \neq j$ for every j : exposed agents are outside U and differ from o , and D has no loops. Every arc $j \rightarrow \sigma(j)$ is an arc of D^+ : an exposure arc if $j \in F$, a need arc otherwise. The agents outside U form a nonempty set (it contains F), so following σ from any of them must repeat an agent, which gives a cycle C of D^+ with $k \geq 2$ distinct agents. Its exposure arcs are arcs $o \rightarrow x_o$ for distinct free agents o , with the distinct labels h_o . Lemma 12 finishes the proof. $\square$

5.4 Proof of the Improvement Lemma

Proof (of Theorem 5). If (P) fails, Lemma 8(b) gives a dominating valid state (a pair chain, or a need cycle). Assume (P). If $F = \emptyset$, then not every agent is in U , so Lemma 11 gives a cycle of D , and Lemma 8(a) gives a dominating valid state. Otherwise $F \neq \emptyset$, and no free agent is a valid absorber. By Lemma 9, no free agent holds nothing. By Lemma 10, $|H_o| \geq |F|$ for every $o \in F$. Lemma 13 gives a dominating valid state.

Each step is a walk in a digraph with at most n vertices, or a greedy choice among at most m goods per free agent, so the dominating state is found in polynomial time. $\square$

Proof (of Corollary 2). Let Y be valid and not dominated by any valid state. If every agent holds its pair, Y is completable (Lemma 11). Otherwise, if no free agent were a valid absorber, Theorem 5 would give a dominating valid state. $\square$

5.5 Remarks

The dichotomy is sharp. Under (P), a free agent holding a good fails as an absorber exactly when $|H_o| \geq |F|$ (Lemma 10), and the count is used only to choose distinct protecting goods. The proof never uses a pair holder as the absorber, except when every agent holds its pair.

The protecting goods cannot be dropped. Even in a Pareto-optimal valid state, every free agent may have an exposed agent, so the protecting goods are needed. Take two agents with rankings $g_0 \succ g_1 \succ g_2$ and $g_1 \succ g_0 \succ g_2$, each holding its top. Both are free, $J = \{g_2\}$, and each is exposed for the other. No valid state dominates this one (each agent's pair contains the other's top), and it is completable only with $H = \{g_2\}$ ([10], k3/simplify/po/hall/NOTES.md, §6).

A second proof. A second, independently found proof of Corollary 2 ([10], k3/simplify/po/potential/NOTES.md) shows that a valid state with no need cycle, pair chain or exchange cycle is completable, by a walk instead of a choice of all representatives at once. It leaves each free agent by an exposure arc whose label has not been used yet (each free agent has at least $|F|$ labels, and fewer have been used), then follows need arcs, in the acyclic D , to a free agent. The walk returns to a free agent it has visited, and the shortest closed part of it is a cycle of D^+ with distinct labels (a *rainbow* cycle). Both proofs first apply pair chains and let an agent holding nothing absorb.

6 The Algorithm Draft and Exchange

6.1 The algorithm

Algorithm 1 lists DE. It receives the values $v_i(g)$; ties among an agent's values are broken by index, as in Section 3.1. The loop of lines 8–20

Algorithm 1 Draft and Exchange (DE)

Require: agents N , goods M , nonnegative values $v_i(g)$, every agent with $|R_i| \leq 3$

- 1: $A \leftarrow N, G \leftarrow M$
- 2: **while** $|A| \geq 2$ and rule R1 applies to some agent of A with respect to G **do** ▷
 peeling
- 3: $i \leftarrow$ the first such agent; $X_i \leftarrow P_i$ (by rule R1)
- 4: $A \leftarrow A \setminus \{i\}, G \leftarrow G \setminus P_i$
- 5: **if** $|A| = 1$ **then** give G to the agent of A and **return** X
- 6: ▷ core case (Lemma 4)
- 7: **for** $i \in A$ in index order **do** $Y_i \leftarrow$ the first of $\{a_i\}, \{b_i\}, \{c_i\}$ whose good is not yet taken, else $\emptyset$
- 8: **loop**
- 9: **if** some $x \notin U$ has $Y_x = \{a_x\}$ and $b_x, c_x \in J$ **then**
- 10: apply the pair chain, or need cycle, of Lemma 8(b)
- 11: **else if** every agent of A is in U **then**
- 12: $o \leftarrow$ the first agent of A ; $H \leftarrow \emptyset$; **break**
- 13: **else if** some $o \in F$ has $Y_o = \emptyset$ **then**
- 14: $o \leftarrow$ the first such agent; $H \leftarrow \emptyset$; **break** ▷ Lemma 9
- 15: **else if** some $o \in F$ has $|H_o| \leq |F| - 1$ **then**
- 16: $o \leftarrow$ the first such agent; $H \leftarrow H_o$; **break** ▷ Lemma 10
- 17: **else if** $F = \emptyset$ **then**
- 18: apply a need cycle, found by following out-arcs of D (Lemma 11)
- 19: **else**
- 20: exchange along the cycle of σ (Lemma 13)
- 21: give each good of H to a different free agent other than o , in index order
- 22: $X_o \leftarrow Y_o \cup (J \setminus H)$; every other $i \in A$ keeps Y_i (plus its good of H , if any)
- 23: **return** X

runs on the core case left by peeling, with the rankings and the state of Section 4.

The loop tests for an absorber only when (P) holds. There the test is a count (Lemma 10). Without (P) the test would ask for a smallest set meeting given sets of one or two goods, which we never need: a pair chain is applied first.

6.2 Correctness and running time

Theorem 6 (DE is correct). *On every instance in which every agent values at most three goods, DE returns a complete EFX₀ allocation in which all bundles but at most one have at most two goods. It peels at most n agents and applies at most $4n$ exchanges (need cycles, pair chains and exchange cycles).*

Proof. Peeling. Each round of the first loop removes one agent, so there are at most n rounds. Let $i_1, \dots, i_p$ be the peeled agents, in order, with sets

$P_1, \dots, P_p$, and let (A_s, G_s) be the agents and goods left after s rounds. Rule R1 for i_s in (A_{s-1}, G_{s-1}) is conditions (i) and (ii) of Lemma 3. So if (A_s, G_s) has an EFX₀ allocation, so has (A_{s-1}, G_{s-1}) , with $X_{i_s} = P_s$ added. By induction from $s = p$ down to $s = 1$, it suffices that the allocation that DE builds for (A_p, G_p) is EFX₀. The loop stops with $|A_p| \geq 1$. If $|A_p| = 1$, the single agent takes G_p and faces no other bundle.

The core. Let $|A_p| \geq 2$. Rule R1 applies to no agent of A_p , so by Lemma 4 every agent of A_p values exactly three goods of G_p and is strictly balanced: the core case, on agents A_p and goods G_p . The draft gives a valid state (Lemma 5). Each branch of the loop that does not stop replaces the state by a valid state that Pareto-dominates it (Lemmas 8, 11 and 13; Lemma 13 applies because the earlier branches did not fire, as in the proof of Theorem 5). So the state stays valid, and the sum of utilities $\sum_{i \in A_p} u_i$, an integer between 0 and $4|A_p|$, grows by at least 1 with each exchange. Hence there are at most $4|A_p| \leq 4n$ exchanges, and the loop stops. When it stops, o is a valid absorber with set H : by Lemma 11 if every agent is in U , by Lemma 9, or by Lemma 10. The last two lines build the completion, which is EFX₀ by Theorem 4, and in which only X_o can have more than two goods.

Shape. Peeled agents receive at most one good. If $|A_p| = 1$, the last agent's bundle is the only one that can be large; otherwise only the absorber's. $\square$

Running time. Each iteration of the loop needs the holder of each good, the needs (at most three goods per agent), the free agents, the sets E_o and H_o , and a walk of at most n steps. Every agent x that holds only its top meets at most one condition among: b_x and c_x both leftover (a pair chain), or exactly one of them leftover and the other held alone by a free agent o (then $x \in E_o$, with h_x the leftover one); so all sets E_o and H_o are found by one pass over the agents. With an array that records the holder of each good, an iteration takes $O(n + m)$ steps, and so does a round of peeling once the relevant goods of each agent are known (it checks rule R1 for each agent and removes the peeled goods). So DE runs in $O(n(n + m))$ steps, reading the input included. This count is machine-checked in Lean, with one unit per elementary operation, array read and array write: at most $750(n + 1)(n + m + 1)$ (Section 8).

Proof (of Theorems 1, 2 and 3). By Theorem 6, DE returns, on every instance in which every agent values at most three goods, a complete EFX₀ allocation in which all bundles but at most one have at most two goods: Theorems 1 and 2 (an agent that values exactly three goods values

at most three). Theorem 6 also bounds the peeling steps by n and the exchanges by $4n$, and each step takes polynomial time, as just discussed: Theorem 3. Every decision of DE compares one value of an agent with another, or with the sum of two of its values (rule R1, the rankings), and the proofs use nothing else about the values; so they hold for nonnegative real values, with the step count of Theorem 3 in the comparison model just described. $\square$

6.3 A worked example

This instance shows that the exchanges of Lemma 13 can need several exposure arcs. On it, every Pareto improvement gives pairs to two agents at once, so a search that applies one short move at a time gets stuck (Section 6.4). Ten goods $g_0, \dots, g_9$ and six agents, in index order $x_1, x'_1, x_2, x'_2, o_1, o_2$, with rankings

agent	$a \succ b \succ c$	draft state Y	after the exchange
x_1	$g_0 \succ g_4 \succ g_6$	$\{g_0\}$ (a)	$\{g_4, g_6\}$ (pair)
x'_1	$g_1 \succ g_4 \succ g_7$	$\{g_1\}$ (a)	$\{g_1\}$ (a)
x_2	$g_2 \succ g_5 \succ g_8$	$\{g_2\}$ (a)	$\{g_5, g_8\}$ (pair)
x'_2	$g_3 \succ g_5 \succ g_9$	$\{g_3\}$ (a)	$\{g_3\}$ (a)
o_1	$g_2 \succ g_3 \succ g_4$	$\{g_4\}$ (c)	$\{g_2\}$ (a)
o_2	$g_0 \succ g_1 \succ g_5$	$\{g_5\}$ (c)	$\{g_0\}$ (a)

Values are any strictly balanced values consistent with these rankings, for instance 4, 3, 2 for every agent.

Peeling and draft. Every agent values exactly three goods and is strictly balanced, so rule R1 applies to nobody, and every good is valued by someone. In the draft, x_1, x'_1, x_2, x'_2 take their tops; o_1 finds g_2 and g_3 taken and takes its c , g_4 ; o_2 takes its c , g_5 . This is the state Y of the table.

The state Y . The needs are $N_{o_1} = \{g_2, g_3\}$ and $N_{o_2} = \{g_0, g_1\}$; the four x 's hold their tops and need nothing. So $NA = \{g_0, g_1, g_2, g_3\}$, all held alone: Y is valid. The leftover goods are $J = \{g_6, g_7, g_8, g_9\}$, $U = \emptyset$, and the free agents are $F = \{o_1, o_2\}$ (their goods g_4, g_5 are not needed). The need arcs are $x_1 \rightarrow o_2$, $x'_1 \rightarrow o_2$, $x_2 \rightarrow o_1$ and $x'_2 \rightarrow o_1$. Condition (P) holds: each x has its b held by an o . For o_1 , the exposed agents are x_1 ($\{b, c\} = \{g_4, g_6\}$) and x'_1 ($\{g_4, g_7\}$), so $H_{o_1} = \{g_6, g_7\}$. Symmetrically $E_{o_2} = \{x_2, x'_2\}$ and $H_{o_2} = \{g_8, g_9\}$. Both have $|H_o| = 2 = |F|$: no free

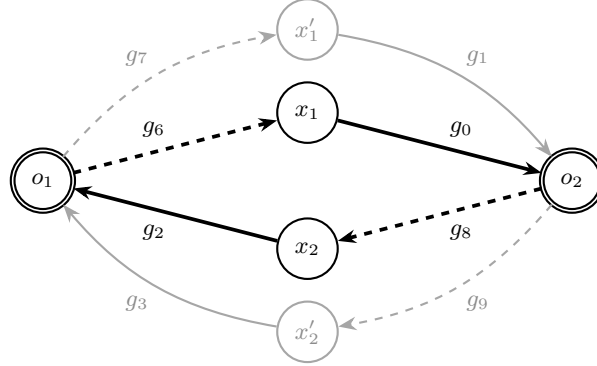


Fig. 1. The exchange digraph D^+ of the draft state of Section 6.3. Double circles are the free agents. Dashed arcs are exposure arcs, labelled by the protecting good; solid arcs are need arcs, labelled by the good that moves. The cycle chosen by DE is drawn in black; along it, x_1 and x_2 take their pairs and o_1 , o_2 take their tops. The other three cycles of D^+ also use two exposure arcs.

agent absorbs. Concretely, if o_1 absorbs, x_1 and x'_1 each need one of their junk goods g_6 , g_7 outside X_{o_1} , but only one other free agent, o_2 , can take one. With $H = \{g_6\}$, say, $X_{o_1} = \{g_4, g_7, g_8, g_9\}$, and x'_1 , which holds g_1 (worth 4), values X_{o_1} without g_8 at $3 + 2 = 5$. In fact all ten candidate completions of Y (absorber o_1 or o_2 , with H empty or one leftover good for the other) fail the EFX_0 test (machine-checked in Lean for the values 4, 3, 2).

No small move helps. D has no cycle (its arcs all end at o_1 or o_2 , which have no out-arcs), and (P) holds, so there is no need cycle and no pair chain. An exchange cycle with one exposure arc $o_1 \rightarrow x$ would have to return to o_1 along need arcs, but every need arc from x_1 or x'_1 ends at o_2 , which has no need arc; the same holds for o_2 . Exhaustive listing shows more: exactly four valid states dominate Y , and each of them gives pairs to two agents (machine-checked in Lean).

The exchange. DE picks representatives greedily: $q_{o_1} = g_6$, with $x_{o_1} = x_1$, and $q_{o_2} = g_8$, with $x_{o_2} = x_2$. The map σ sends $o_1 \mapsto x_1$, $x_1 \mapsto o_2$ (o_2 needs g_0), $o_2 \mapsto x_2$ and $x_2 \mapsto o_1$ (o_1 needs g_2), giving the cycle of Figure 1. Along it, x_1 takes $\{g_4, g_6\}$, o_2 takes g_0 , x_2 takes $\{g_5, g_8\}$ and o_1 takes g_2 : two new pairs in one exchange. The sum of utilities rises from 14 to 20.

Completion. In the new state nobody needs anything ($NA = \emptyset$), the leftover goods are g_7 and g_9 , and the free agents are x'_1 , x'_2 , o_1 , o_2 . Condition

(P) holds (b of x'_1 is g_4 , now held by x_1 ; similarly for the others). The first free agent, x'_1 , has nobody exposed: the other agents holding only their tops are x'_2 , o_1 and o_2 , and none has both b and c in $J \cup \{g_1\} = \{g_1, g_7, g_9\}$. So x'_1 absorbs with $H = \emptyset$, and DE returns

$$\begin{aligned} X_{x_1} &= \{g_4, g_6\}, & X_{x'_1} &= \{g_1, g_7, g_9\}, & X_{x_2} &= \{g_5, g_8\}, \\ X_{x'_2} &= \{g_3\}, & X_{o_1} &= \{g_2\}, & X_{o_2} &= \{g_0\}. \end{aligned}$$

It is EFX₀ by Theorem 4. Directly, with values 4, 3, 2: the only bundle of three goods, $X_{x'_1}$, is worth 3 to o_2 (only g_1), which holds g_0 , worth 4, and 2 to x'_2 (only g_9), which holds g_3 , worth 4; the bundles of two goods are worth at most 3 to any other agent after removing one good, and every other agent holds at least 3 (a top worth 4, or a pair worth 5, or $X_{x'_1}$ worth 6). Our script checks this for 2,000 random strictly balanced valuations consistent with the rankings (Section 8).

6.4 How large must an exchange be?

One exposure arc is not enough. A natural simpler algorithm would end with at most one *rotation*: an agent k exposed for a candidate absorber r takes its pair, and its top travels along a chain of agents that need it, ending at r . This is an exchange cycle $r \rightarrow k \rightarrow \dots \rightarrow r$ with one exposure arc (or a pair chain when b_k, c_k are leftover). After an arbitrary draft one rotation does not suffice: in Section 6.3 every improvement gives pairs to two agents, so a local search that applies need cycles, pair chains and one-arc rotations gets stuck there in a state that is not completable.

Call a *short move* a need cycle, a pair chain, or an exchange along a cycle of D^+ with exactly one exposure arc. The next proposition, Lemma 3 of the second proof's notes [10], machine-checked in Lean, shows that the instance of Section 6.3 is as small as possible in its number of agents.

Proposition 1 (when short moves are not enough). *In the core case, let Y be a valid state in which no free agent is a valid absorber and some agent is not a pair holder, and to which no short move applies. Then $|F| \geq 2$, $n \geq 6$ and $m \geq 8$; if $|F| \geq 3$, then $n \geq 9$ and $m \geq 12$. Both bounds $n = 6$ and $m = 8$ are attained.*

Proof. No pair chain applies, so (P) holds (Lemma 8(b)). No need cycle applies, so D is acyclic, and every walk along need arcs ends at an agent with no out-arc, which is free by (F2). By Lemma 11, $F \neq \emptyset$. By Lemma 9, every free agent holds a good, and by Lemma 10, $|H_o| \geq |F|$ for every

$o \in F$. Every agent holds at least one good: pair holders hold two, other agents outside F hold a needed good, and free agents hold a good. Let $k = |F|$.

If $k = 1$, let $F = \{o\}$. As $|H_o| \geq 1$, some $x \in E_o$ exists, and $x \neq o$, so x is not free. The walk along need arcs from x ends at a free agent, that is, at o . Since D is acyclic, $o \rightarrow x \rightarrow \dots \rightarrow o$ is a cycle of D^+ with one exposure arc: a short move (Lemma 12). So $k \geq 2$.

The sets E_o , $o \in F$, are pairwise disjoint (the remark after Lemma 10), and $|E_o| \geq |H_o| \geq k$.

If $k = 2$, let $F = \{o_1, o_2\}$ and $x \in E_{o_1}$. The walk along need arcs from x cannot end at o_1 , since that would give a short move as above; so it ends at o_2 . As $|E_{o_1}| \geq 2$, some $x \in E_{o_1}$ differs from o_2 ; this x is not free, so its walk enters o_2 by a need arc, and o_2 needs a good. Then o_2 does not hold only its top, so $o_2 \notin E_{o_1}$, and E_{o_1} consists of at least two agents outside F . Symmetrically for E_{o_2} . So $n \geq 2 + 2 + 2 = 6$. At least n goods are held and $|J| \geq |H_{o_1}| \geq 2$, so $m \geq 8$.

If $k \geq 3$, the disjoint sets E_o contain at least k^2 agents, of which at most k are free, so $n \geq k + (k^2 - k) = k^2 \geq 9$, and $m \geq n + |J| \geq k^2 + k \geq 12$.

Attained. The instance of Section 6.3 has $n = 6$ (and $m = 10$). An instance with $n = 6$ and $m = 8$ is: o_1 ranks $g_4 \succ g_5 \succ g_0$, o_2 ranks $g_6 \succ g_7 \succ g_1$, x_1 ranks $g_4 \succ g_1 \succ g_2$, x'_1 ranks $g_5 \succ g_3 \succ g_1$, x_2 ranks $g_6 \succ g_0 \succ g_2$, x'_2 ranks $g_7 \succ g_3 \succ g_0$, with o_1 holding g_0 , o_2 holding g_1 , and every x holding its top. Then $J = \{g_2, g_3\}$, $F = \{o_1, o_2\}$, $E_{o_2} = \{x_1, x'_1\}$ with protecting goods g_2, g_3 , and $E_{o_1} = \{x_2, x'_2\}$ with protecting goods g_2, g_3 ; no free agent absorbs, and no short move applies (checked by our script and in Lean). $\square$

In the second instance the distinct labels matter. Of the four cycles $o_2 \rightarrow x \rightarrow o_1 \rightarrow x' \rightarrow o_2$ with two exposure arcs, two use the same protecting good twice, and exchanging along them would give g_2 (or g_3) to two agents; the other two give valid dominating states (checked by our script and in Lean).

No bounded number of exposure arcs suffices. The second proof's notes [10] (§7) give valid states with k free agents in a ring: o_i needs the goods at the roots of a binary tree of agents that need their children's goods, and the leaves of the tree of o_{i+1} hold their tops and are exposed for o_i . With trees of depth d , every cycle of D^+ , with distinct labels or not, uses exactly k exposure arcs; and no free agent absorbs if and only if $k \leq 2^d$, the number of leaves of each tree (for $k > 2^d$,

the agent o_0 absorbs). Both are machine-checked in Lean for every $k \geq 2$ and $d \geq 1$, with private or shared labels. The notes checked $(k, \text{depth}) \in \{(2, 1), (3, 2), (4, 2), (5, 3)\}$, which satisfy $k \leq 2^d$; the referee ran DE on rings with $k \leq 9$. We have not proved that every Pareto improvement there upgrades k agents.

7 Limits of the Shape

Could every instance have an EFX₀ allocation with all bundles of at most two goods, or with one bundle of exactly three? No. The proof uses the full description of safety for agents with distinct values.

Lemma 14 (cases of safety). *Let agent i have $R_i = \{a, b, c\}$ with $v_i(a) > v_i(b) > v_i(c) > 0$ and $v_i(a) < v_i(b) + v_i(c)$, and let X be an allocation. Then i is safe in X if and only if one of the following holds:*

- (T) *i holds a , and either i also holds b or c , or b and c are not together in a bundle of at least three goods;*
- (BC) *i holds b and c ;*
- (B) *i holds b , and a is alone;*
- (C) *i holds c , and a and b are alone;*
- (E) *a , b and c are all alone.*

Proof. Write a , b , c also for $v_i(a)$, $v_i(b)$, $v_i(c)$. The sums of subsets of $\{a, b, c\}$ are ordered as $0 < c < b < a < b + c < a + c < a + b < a + b + c$: $c < b < a$ by assumption, $a < b + c$ by strict balance, $b + c < a + c$ since $b < a$, $a + c < a + b$ since $c < b$, and $a + b < a + b + c$ since $c > 0$. Let $S = X_i \cap R_i$, so $v_i(X_i) = v_i(S)$. For another bundle X_j , let $T = X_j \cap R_i$. By Lemma 1, $\theta_i(X_j) = 0$ if $T = \emptyset$ or $|X_j| = 1$; $\theta_i(X_j) = v_i(T)$ if X_j has a good outside R_i ; and if $X_j = T$ has two goods, $\theta_i(X_j)$ is the larger value. We go through the possible sets S .

- $S \supseteq \{a, b\}$ or $S \supseteq \{a, c\}$. Then $v_i(S) \geq a + c$, and every other bundle meets R_i in at most one good, so its threat is at most $b < a + c$. Safe; this is (T).
- $S = \{b, c\}$. Only a lies outside S , so every threat is at most $a < b + c$. Safe; this is (BC).
- $S = \{a\}$. If b and c lie in different bundles, each threat is at most $b < a$. If they lie in the same bundle X_j and $X_j = \{b, c\}$, the threat is $b < a$. If $X_j \supsetneq \{b, c\}$, then X_j contains a good outside R_i (the good a is i 's), so $|X_j| \geq 3$ and the threat is $b + c > a$. So i is safe if and only if b and c are not together in a bundle of at least three goods: (T).

- $S = \{b\}$. If a is not alone, its bundle is $\{a, c\}$, with threat a , or contains a good outside R_i , with threat at least a ; either way the threat exceeds b . If a is alone, its threat is 0, and the bundle of c has threat at most $c < b$. So i is safe if and only if a is alone: (B).
- $S = \{c\}$. As before, a must be alone. If b is not alone, its bundle is $\{a, b\}$ (threat a) or contains a good outside R_i (threat at least b); either way the threat exceeds c . If a and b are alone, every threat is 0. So i is safe if and only if a and b are alone: (C).
- $S = \emptyset$. Then $v_i(S) = 0$, and a bundle with a good of R_i and at least two goods has a positive threat. So i is safe if and only if a , b and c are all alone: (E).

Conversely, each of (T), (BC), (B), (C), (E) describes one of the safe situations above: (T) is the first and third items; (BC) is $S = \{b, c\}$ or $S = \{a, b, c\}$; (B) forces $a \notin S$, so $S = \{b\}$ with a alone, or $S = \{b, c\}$; (C) forces $S = \{c\}$ with a and b alone; and (E) means that S is empty or a single alone good, which is (T), (B) or (C). $\square$

Proposition 2 (limits of the shape). *Take values with $v_i(a_i) > v_i(b_i) > v_i(c_i) > 0$ and $v_i(a_i) < v_i(b_i) + v_i(c_i)$ for every agent.*

- (a) *Let agents 0, 1, 2 rank $g_0 \succ g_1 \succ p_i$, where p_i is valued only by agent i . No EFX_0 allocation has all bundles of at most two goods, and $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$, to the three agents in any order, is EFX_0 .*
- (b) *Let agents 0, 1, 2, 3 rank $g_0 \succ g_2 \succ p_0, g_0 \succ g_2 \succ p_1, g_1 \succ g_2 \succ p_2$ and $g_1 \succ g_2 \succ p_3$, with p_i valued only by agent i . Every EFX_0 allocation has bundle sizes 4, 1, 1, 1, and one exists.*

Proof. (a) Five goods in three bundles of at most two goods have sizes 2, 2, 1, so exactly one good is alone. Good g_0 is everyone's top and has one holder, so two agents do not hold their top: case (T) of Lemma 14 fails for them, and cases (C) and (E) need two or three alone goods. So each of these two agents is in case (BC) or (B), and both hold g_1 , which is impossible. In $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$, the holder of g_0 is in case (T), the holder of g_1 in case (B), and the third agent in case (C).

(b) Each of the tops g_0 and g_1 has an agent ranking it first that does not hold it, a *loser*; the two losers are different agents. A loser is safe only by (BC) or (B), which need it to hold g_2 , or by (C) or (E), which need g_2 alone. The two losers cannot both hold g_2 , so g_2 is alone, and then no agent can be in case (BC), which would put g_2 with another good. A loser holding $\{g_2\}$ is in case (B) or (E) and needs its top alone; a loser not holding g_2 is in case (C) or (E) and needs its top and g_2 alone. So g_0 ,

Table 2. The claims of this paper and their evidence. *Written*: proved in this paper. *Lean*: machine-checked in Lean 4 over the natural numbers (`lean/EFX/K3DE*.lean` in [10]). *Refereed*: one independent referee report, by an AI agent (Section 8), with no error found.

Claim	Evidence
Lemmas 1–4, Cor. 1	written; Lean
Example 1; protecting goods cannot be dropped (Section 5.5)	Lean
Thm. 5 (Improvement Lemma), Lemmas 6–13	written; Lean (Lemmas 8(a), 11 and 13 inside the theorems on DE’s step); derived twice; refereed; checked by computer (three implementations)
Thm. 4 (soundness)	written; Lean; the same argument refereed once in an earlier form of the notes [10]; raw EFX ₀ checks
Thms. 1, 2, 3, 6 (DE)	written; Lean, also on ordered values, the running time included; the referee and our script ran DE, with peeling, on random instances
Prop. 1 (short moves); the facts of Section 6.3	written; Lean; also checked by our script
Lem. 14, Prop. 2 (limits)	written; Lean (every strictly balanced valuation with distinct values); every allocation listed by our script
Rings need k exposure arcs	Lean (every k and depth d ; no free agent absorbs iff $k \leq 2^d$)

g_1 and g_2 are alone, held by three different agents, and the fourth bundle is $\{p_0, p_1, p_2, p_3\}$. Such an allocation exists: g_0 to agent 0, g_1 to agent 2, g_2 to agent 1 and the four goods p_i to agent 3 is EFX₀ (cases T, B, T, C). $\square$

Lemma 14 is machine-checked in Lean (over the natural numbers), and so are both parts, for every valuation with the stated strict conditions and also for ordered values, by evaluating the cases of Lemma 14 on every allocation; our script confirms them for several valuations. In (a) every agent values exactly three goods and is strictly balanced, so it is a core-case instance with no EFX₀ allocation of bundles of at most two goods: the one large bundle of Theorem 2 cannot be avoided. By (b), its size cannot be bounded by three.

8 Verification Status and Evidence

What is proved, and how it was checked. The Improvement Lemma, the soundness theorem and the correctness of DE are written proofs. They have been checked by computer as described in this section, and they are machine-checked in Lean (the last paragraphs of this section). The Improvement Lemma was found twice, independently: the proof of Section 5 (`k3/simplify/po/hall/NOTES.md` in [10]) and the rainbow walk of Section 5.5 (`k3/simplify/po/potential/NOTES.md`). An independent referee read the first proof and found no error (`k3/simplify/po/referee/README.md`). All three, the two provers and the referee, were AI agents: separate sessions of the coding assistant (Claude Code) that helped with this work, the referee given only the written proof. There has been no human peer review. Its four presentation fixes are applied here: the explicit completion and the proof of the owner constraint (Theorem 4); the remark that h_x depends only on x (after Lemma 10); the explicit use of Lemma 8(a) when $F = \emptyset$; and the wording of the tests (one completion checked per state). The soundness argument was refereed once, with no error, in an earlier form of the notes (`proofs/k3s_standalone.md`, §5).

Three implementations. The tests are evidence, not proof. In each, every claim of the proof is an assertion, every exchange is re-checked for validity and Pareto domination, and allocations are checked against the raw definition of EFX_0 with exact arithmetic and values 4, 3, 2 (for distinct, strictly balanced values EFX_0 depends only on the rankings, by Lemma 14). “Every profile” means every assignment of three ranked goods to each agent, up to renaming goods. No test found a failure.

1. *The first proof as code* (`k3/simplify/po/hall/hall.py`): every valid state of every profile with $n = 2$ ($m \leq 6$), $n = 3$ ($m \leq 7$) and $n = 4$, $m = 5$, that is, 1,693,750 valid states, of which the 269,028 that are not completable were each improved as the proof prescribes; 183,484 further states from random profiles ($n \leq 8$) and cores ($n = 5, 6$); for each completable state, the completion it built (one per state). Its generator of random states cannot produce states whose need digraph has a cycle; the referee’s rejection-sampled states below can; all 774,248 Pareto-optimal valid states of the exhaustive sets and of 3,000 random profiles; and DE’s loop on 44,971 random and core profiles.
2. *The second proof as code* (`k3/simplify/po/potential/exchange.py`): about 15.5 million valid states, from every profile with $n = 2$ ($m \leq 6$), $n = 3$ ($m \leq 7$) and $n = 4$ ($m \leq 6$), 100,000 random profiles with $n \leq 8$, and samples of cores with $n = 5, 6$.

3. *The referee's implementation* (`k3/simplify/po/referee/`), written from the text of the first proof alone, with random choices wherever the proof leaves one; about 2.9 million runs: DE's loop on every profile with $n = 2$ ($m \leq 6$), $n = 3$ ($m \leq 7$) and $n = 4$ ($m = 5, 6$), 2,006,886 profiles, on 30,700 core profiles with $n = 5$ and on 200,000 random profiles with $n \leq 9$; 273,805 valid states, every non-completable one improved once (115,304 improvement runs in all, counted as in its log); soundness on 1,130,775 completions (all completions of all valid states with $n \leq 3$, and of sampled states with $n = 4$); 65,816 random valid states sampled by rejection, including states whose need digraph has a cycle; 3,000 rings with up to 9 free agents and 304 agents; and peeling by R1 followed by DE on 300,000 instances with real values.

The script of this paper. `paper/k3-simple/examples/check_examples.py`, a fourth implementation written from this paper, recomputes every example and number of Sections 1, 6.3, 6.4 and 7, cross-checks Section 6.3 and the first rows of item 1 with `hall.py`, and runs DE, peeling included, on 40,000 random instances: 20,000 with $n \leq 8$ and 0 to 3 valued goods per agent, values in $\{1, \dots, 6\}$ (ties, agents with $v(a) \geq v(b) + v(c)$, goods nobody values), and 20,000 in which every agent values three goods and is strictly balanced ($n \leq 9$). Every output was complete and EFX₀, with at most one bundle of more than two goods; 10,523 runs needed an exchange, at most 8. The script exits with status 0, which it does only if every check passes.

The Lean formalization. The files `lean/EFX/K3DE.lean`, `K3DEImprove.lean`, `K3DEAlgo.lean` and `K3DEExamples.lean` of [10] formalize Sections 4 and 5, Algorithm 1 and Theorem 6 in core Lean 4: no library beyond Lean's own, no unproved step, every declaration on Lean's three standard axioms, all checked by the repository's `lean/check.sh`, which also replays the library in a fresh kernel. States are the valid pre-allocations of the repository's Lean library (`lean/EFX/PreAlloc.lean`: an agent holds nothing, one of its goods, or its pair, with needs given by the rankings), and soundness reuses its theorem on completions. The Improvement Lemma is proved for a computable step that follows the loop of Algorithm 1. One lemma covers need cycles, pair chains and exchange cycles: a pair chain is closed by an arc from its last agent, which is free, back to the agent that takes its pair. The cycle is the one through $\sigma^n(s)$, where s is the agent x at which (P) fails for a pair chain, and the first agent outside U otherwise; a theorem states that every move is an exchange along such a cycle. The lemma holds for any

rankings of three distinct goods per agent; values enter only through soundness. DE itself is a Lean function. The main theorem states that, on every instance with natural-number values in which every agent positively values at most three goods, DE returns an EFX_0 allocation in which all bundles but at most one have at most two goods, after at most $4n$ exchanges. Evaluation by Lean’s kernel reproduces the output of Section 6.3. The formal definitions and statements were written by the same AI assistant that helped with this work, following this paper; independent AI referee sessions compared them with this text and found no mismatch (`k3/simplify/po/referee/lean_audit.md`, `lean_audit2.md` and `lean_audit3.md` in [10]); its documentation fixes are applied.

Seven further files, from `K3DEPrelim.lean` to `K3DERemarks.lean`, formalize the rest, with the same rules: the lemmas of Section 3; Lemma 14 and Proposition 2 for every valuation with distinct, strictly balanced values; Proposition 1, with both instances; Example 1; the remark that the protecting goods cannot be dropped; the facts of Section 6.3, Figure 1 included; the rings of Section 6.4 for every k and depth; the bound of n peeling rounds; and DE on ordered values (such as the nonnegative reals): it runs on natural-number values computed from the given ones with at most $n(m+12)$ comparisons, which answer every comparison of two subset sums of an agent as the given values do (`lean/EFX/RealValues.lean`), so its output is EFX_0 for the given values. These files were written by separate AI agent sessions of the same assistant and checked by the coordinating session against this text.

Five more files, from `K3DECost.lean` to `K3DECostReal.lean`, formalize the running time. A counted program, proved to return exactly DE’s output on every instance, takes at most $750(n+1)(n+m+1)$ units, reading the input included: one unit per elementary operation, comparison, array read and array write, as in a random-access machine, with arrays used single-threaded (a fresh copy, charged, when an old version is still needed). Each loop iteration takes at most $175n + 11m + 8$ units (one of them for the exchange counter), and each peeling round at most $30n + m + 3$. On ordered values, computing the natural-number values takes at most $n(m+12)$ comparisons and $10nm + 971n$ other units. That the units match a machine is a property of the counted program, checked by reading it.

What is not done. On ordered values, DE decides on the natural-number values; that each of its decisions is one comparison of the given values is argued in Section 6.2, not formalized (the formal count is that of comput-

ing those values plus DE on them, also $O(n(n + m))$). The rainbow-walk proof is not formalized. The Improvement Lemma has had one referee.

9 Conclusion and Open Problems

Every instance in which each agent positively values at most three goods has a complete EFX₀ allocation with at most one bundle of more than two goods, and DE computes one with at most n peeling steps and $4n$ exchanges. The proof rests on one lemma: a valid state that no free agent can complete is Pareto-dominated by a valid state, through a need cycle, a pair chain, or an exchange cycle whose exposure arcs have distinct protecting goods, which a greedy choice of distinct representatives provides.

Open problems. (1) EFX₀ with four relevant goods per agent (for ordinary EFX, see [15]). With four goods an agent can hold one, two or three of them, and the order of holdings is not determined by the ranking alone, so the utilities of Section 4 have no direct analogue; a single rotation already fails for $n = 5$ with one agent valuing four goods [10]. (2) Other potentials: the proof works for any potential that grows under Pareto improvements among valid states; minimizing $|NA|$ alone gave completable states in every test of [10], unproved. (3) A better bound than $4n$ exchanges (on random instances at most 8 were needed; on the rings many more), or a draft that needs none. (4) The least size of the large bundle; Proposition 2(b) shows that three goods do not always suffice.

Acknowledgments. The proofs, code and the Lean formalizations were developed with the assistance of AI coding agents (Claude Code); this paper was also drafted with their assistance.

References

1. Afshinmehr, M., Ashuri, A., Mahmoudkhan, P., Mehlhorn, K., Shahrezaei, A.M.: EFX allocations exist on multi-graphs. arXiv preprint arXiv:2606.18665 (2026), accepted at the 27th ACM Conference on Economics and Computation (EC 2026)
2. Akrami, H., Mayorov, A., Mehlhorn, K., Srinivas, S., Weidenbach, C.: A counterexample to EFX $n \geq 3$ agents, $m \geq n + 5$ items, submodular valuations via SAT-solving. arXiv preprint arXiv:2604.18216 (2026)
3. Alkassar, E., Fouz, M., Mehlhorn, K.: Complete EFX allocations exist for four additive agents and up to nine goods. arXiv preprint arXiv:2608.08590 (2026)
4. Amanatidis, G., Birmpas, G., Filos-Ratsikas, A., Voudouris, A.A.: Fair division of indivisible goods: A survey. In: Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22). pp. 5385–5393 (2022). <https://doi.org/10.24963/ijcai.2022/756>, arXiv:2202.07551

5. Chaudhury, B.R., Garg, J., Mehlhorn, K.: EFX exists for three agents. *Journal of the ACM* **71**(1), 4:1–4:27 (2024). <https://doi.org/10.1145/3616009>, conference version in Proc. 21st ACM Conference on Economics and Computation (EC 2020), pp. 1–19; full version arXiv:2002.05119
6. Christodoulou, G., Fiat, A., Koutsoupias, E., Sgouritsa, A.: Envy-freeness up to any good allocations on graphs. *Mathematics of Operations Research* (2026). <https://doi.org/10.1287/moor.2025.0915>, conference version: “Fair Allocation in Graphs”, in Proc. 24th ACM Conference on Economics and Computation (EC 2023), pp. 473–488
7. HV, V.P., Ghosal, P., Nimbhorkar, P., Varma, N.: EFX exists for three types of agents. In: *Proceedings of the 26th ACM Conference on Economics and Computation (EC 2025)*. pp. 101–128. ACM (2025). <https://doi.org/10.1145/3736252.3742509>, arXiv:2410.13580
8. Kaviani, A., Seddighin, M., Shahrezaei, A.: Almost envy-free allocation of indivisible goods: A tale of two valuations. In: *Web and Internet Economics (WINE 2024)*. pp. 261–276. *Lecture Notes in Computer Science*, Springer (2026). https://doi.org/10.1007/978-3-032-08560-3_15, full version arXiv:2407.05139
9. Lianes, T., Sgouritsa, A., Sotiriou, M.M.: EFX allocation in (multi)hypergraphs. *Proceedings of the AAAI Conference on Artificial Intelligence* **40**(20), 17102–17110 (2026). <https://doi.org/10.1609/aaai.v40i20.38759>, AAAI-26; arXiv:2608.03171
10. Lin, E.: efx0-three-goods: EFX₀ with at most three relevant goods per agent (notes, tests, referee reports and Lean formalization). GitHub repository, commit 462c724, <https://github.com/evanlin23/efx0-three-goods> (2026)
11. Lin, E., Osterhaus, G.: Strong EFX allocations for 2-relevant agents: Lean 4 formalization. GitHub repository `mr-d-efx`, tag `v1.2.0` (commit `fa42aa1`, 23 September 2026), <https://github.com/evanlin23/mr-d-efx> (2026)
12. Mackenzie, S., Suzuki, M.: Counterexamples to EFX for submodular and subadditive valuations. arXiv preprint arXiv:2605.06451 (2026)
13. Mahara, R.: Extension of additive valuations to general valuations on the existence of EFX. *Mathematics of Operations Research* **49**(2), 1263–1277 (2024). <https://doi.org/10.1287/moor.2022.0044>, conference version in Proc. 29th Annual European Symposium on Algorithms (ESA 2021), LIPIcs 204, 66:1–66:15; arXiv:2107.09901
14. Sgouritsa, A., Sotiriou, M.M.: On the existence of EFX allocations in multigraphs. arXiv preprint arXiv:2502.09777 (2025), extended abstract, without the bipartite case, in Proc. 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), pp. 2735–2737
15. Viswanathan, M., Mehta, R.: On the existence of EFX under picky or non-differentiative agents. In: *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)*. pp. 2534–2536 (2024). <https://doi.org/10.65109/osos1146>, extended abstract